Study Studio — Proposed Pricing Structure

Credit prices and plan economics · 9 October 2026 · Study Studio only (OctoNotes is priced separately) · companion to the AI Cost Sheet of 8 October 2026

1. Summary

The draft plans from Finance (Free / Pro / Premium, their prices and their monthly credit allowances) are kept as they are. What changes is **how many credits each AI action costs**, and the rule we use to set those prices.

- **Price every action in credits from one rule:** even if a user spends every credit on the most expensive action, AI must cost at most **30% of the money we receive**. The other 70% pays for servers, free users and profit.
- **Proposed prices:** Ask AI **20** · Standard build **150** · Deep build **200** · Concept map **50** · Recall grading **free**.
- **Every paid plan keeps 64-88% of its net revenue** after the worst-case AI bill, before servers. At typical use (about 30% of the allowance) AI is 3-9% of revenue.
- **Replace the line "100 OctoCredits = about $1 of AI cost".** Measured, an AI action costs $0.009-0.09, not $1. If a developer implemented that rate literally (1 credit = 1 cent of AI), every plan would lose money.
- **Replace "cost + 10%, rounded up to the next dollar, minimum 100 credits".** It only works because of the minimum, it makes every action cost the same 100 credits, and it gives a Free user 5 actions a month.
- **Remove "larger OctoNotes/transcription usage" from Premium** in this sheet; OctoNotes gets its own pricing.

2. What the current draft does, and why it needs to change

The draft sets prices by cost-plus: run a feature 10 times, take the average AI cost, add 10%, round up to the next whole dollar, and charge 100 credits per dollar, with a minimum of 100 credits per AI action.

Draft rule	What actually happens	Fix
100 credits = about $1 of AI cost	An Ask AI answer costs $0.0092; a full studio build $0.036. 100 credits buy 1-9 cents of AI, not $1.	Drop the table. A credit's value comes from what users pay for it (section 4).
Cost + 10%, round up to the next dollar	Every action costs well under $1, so every action rounds up to $1 = 100 credits. The real markup is 10-100x, not 10%.	Fixed credit price per action, set from the 30% rule (section 5).
Minimum 100 credits per action	Profitable, but Free = 5 actions a month and Pro = 20. Recall grading ($0.00005) would cost 100 credits per spoken answer.	Small actions cost few credits; grading is free.
Premium: larger OctoNotes/transcription usage	Not part of Study Studio; transcription is billed per audio minute and is not measured here.	Remove from this sheet; price OctoNotes separately.
Promo +1,000 credits on Pro	Doubles Pro's allowance and its worst-case AI bill for as long as it runs.	Give it for the first 3 months only (section 8).

Under the draft as written we do **not** lose money on the Study Studio features measured so far, but only because of the 100-credit minimum. The stated reasoning ("about $1 of AI") is wrong by 10-100x, and anyone pricing a new feature from it would price it wrongly.

3. Measured AI cost per action

From the production benchmark (AI Cost Sheet, 8 October 2026): real studios built in the live app with thinking off, cost taken from OpenRouter's own usage reports. "Large" is an upload big enough to fill the 120,000-character outline limit — the most a single build can cost today.

Action	Model	Typical cost	Large / worst
Ask AI (one answer)	Claude Sonnet 4.6	$0.0092	$0.0092
Standard build (one studio)	Gemini 3.1 Flash Lite	$0.0357	$0.0870
Deep build (one studio)	GLM 5.3 Flash	$0.0367	$0.0790
Concept map (one lesson)	Gemini 3.1 Flash Lite	$0.0170	$0.0250
Recall grading (one spoken answer)	Gemini 3.1 Flash Lite	$0.00005	$0.00005

Ingesting uploads (text extraction, OCR, embeddings) runs on our own server and has no per-call AI cost; it is part of server cost.

4. What one credit is worth

A credit is worth what the user paid for it, after payment fees, divided by the credits they get. Plan credits are much cheaper than pack credits, so prices must be safe at the **cheapest** credit we sell.

Fees assumed: Stripe card 2.9% + $0.30 per charge, plus Stripe Billing 0.7% on subscriptions. Pro counts its 1,000 promotional credits.

Plan	Billing	Price	Net per month	Credits / month	Value of 1 credit
Pro	monthly (launch)	$4.99/mo	$4.51	2,000	$0.00226
Pro	yearly (launch)	$49.99/yr	$3.99	2,000	$0.00200
Pro	monthly (later)	$9.99/mo	$9.33	2,000	$0.00467
Premium	monthly (launch)	$9.99/mo	$9.33	5,000	$0.00187
Premium	yearly (launch)	$99.99/yr	$8.01	5,000	$0.00160
Premium	monthly (later)	$19.99/mo	$18.97	5,000	$0.00379
Credit pack	one-off	$9.99	$9.40 (once)	1,000	$0.00940
Credit pack	one-off	$39.99	$38.53 (once)	5,000	$0.00771
Free	—	$0	$0	500	$0 (we pay)

The cheapest credit is Premium yearly at about **$0.00160**. A pack credit is worth about 5x more, which is why prices set from pack value (as in the first cost sheet) would lose money on heavy plan users.

5. The pricing rule

Worst-case AI cost may use at most 30% of the value of the cheapest credit:

Credits per action = largest AI cost of the action ÷ (30% × $0.00160) = largest AI cost ÷ $0.00048

30% is a common ceiling for the AI share of revenue in AI apps. It leaves 70% for servers, free users, app-store fees and profit, even for a user who spends everything on the most expensive action. The raw result is rounded to a round number; where the typical cost is far below the large one (builds), the price sits between the two so typical studios are not overcharged. Deep build is priced above the rule on purpose: it is the premium option (2-3x more lessons for a longer wait), so it costs more than a Standard build.

Action	Largest AI cost	Rule gives	Proposed	AI share at typical cost	AI share at largest cost
Ask AI (one answer)	$0.0092	19	**20**	29%	29%
Standard build (one studio)	$0.0870	181	**150**	15%	36%
Deep build (one studio)	$0.0790	164	**200**	11%	25%
Concept map (one lesson)	$0.0250	52	**50**	21%	31%
Recall grading (one spoken answer)	$0.00005	0.1	**free**	—	—

AI share is the AI cost divided by what those credits are worth on the cheapest plan (Premium yearly). On every other plan the share is lower. Only a user who builds nothing but very large studios exceeds 30% — and still stays under 40%.

6. Proposed credit price list

Action	Credits	Setting (system_settings)	Notes
Ask AI (one answer)	**20**	study_price_ask	Claude Sonnet 4.6 today
Standard build (one studio)	**150**	study_price_build	Gemini 3.1 Flash Lite, ~20 s
Deep build (one studio)	**200**	study_price_build_deep *	GLM 5.3 Flash, 6-7 min, 2-3x more lessons
Concept map (one lesson)	**50**	study_price_concepts	older studios only; new builds include it
Recall grading (one spoken answer)	**free (0)**	study_price_grade	free: would drain a recall session otherwise

Prices are read by the server from `system_settings` once a minute — the admin team can change them at any time without a release. A user without enough credits gets a clear "not enough credits" message before any AI call is made, and a retried request is never charged twice. * **Deep build** needs one small change before launch: a Standard/Deep choice in the app and its own price key.

What a month's allowance buys

Plan	Credits	Ask AI answers	Standard builds	Deep builds	A realistic month
Free	500	25	3	2	2 studios + 5 questions
Pro (launch, with promo)	2,000	100	13	10	6 studios + 55 questions
Pro (after promo)	1,000	50	6	5	3 studios + 27 questions
Premium	5,000	250	33	25	15 studios + 130 questions

Recall grading, flashcards, quizzes and Lightning Recall on an existing studio use no credits.

7. Plan economics: AI cost against revenue

Worst case: the user spends every credit on the action with the highest AI cost per credit (large Standard builds, $0.00058 per credit). **Typical**: 30% of the allowance used, mostly Ask AI ($0.00046 per credit). Server cost is not included yet — see section 8.

Plan	Billing	Net / month	AI worst case	AI share	Left (worst)	AI typical	Left (typical)
Pro	monthly (launch)	$4.51	$1.16	26%	**+$3.35**	$0.27	**+$4.24**
Pro	yearly (launch)	$3.99	$1.16	29%	**+$2.83**	$0.27	**+$3.72**
Pro	monthly (later)	$9.33	$1.16	12%	**+$8.17**	$0.27	**+$9.06**
Premium	monthly (launch)	$9.33	$2.90	31%	**+$6.43**	$0.69	**+$8.64**
Premium	yearly (launch)	$8.01	$2.90	36%	**+$5.11**	$0.69	**+$7.32**
Premium	monthly (later)	$18.97	$2.90	15%	**+$16.07**	$0.69	**+$18.28**
Free	—	$0	$0.29	—	**-$0.29**	$0.07	**-$0.07**

Every paid plan is profitable on AI in the worst case. The thinnest plan is Premium yearly: about 29% if the user only asks questions, about 36% if they only build very large studios. The "later" prices (Pro $9.99, Premium $19.99) roughly halve the AI share.

8. Servers, free users and profit

What is left after AI must also pay for the servers and for the AI that free users consume:

Profit per paying user = net revenue - AI cost - (server cost ÷ paying users) - (free users per paying user × AI cost per free user)

Server cost is mostly fixed (the VPS, database, storage, backups), so its share per user falls as users grow. Free users are the main risk: each one can use up to $0.29 of AI a month, and there are usually many free users for every paying one.

Example: Pro monthly at launch ($4.51 net), server **$60/month as a placeholder** shared by 100 paying users ($0.60 each). Replace the placeholder with the real server bill.

Free users per paying user	Usage	AI (paying user)	Free users' AI	Server share	Profit / paying user	Margin
5	typical (30%, Ask)	$0.27	$0.34	$0.60	**+$3.29**	73%
5	worst (100%, large builds)	$1.16	$1.45	$0.60	**+$1.30**	29%
10	typical (30%, Ask)	$0.27	$0.69	$0.60	**+$2.95**	65%
10	worst (100%, large builds)	$1.16	$2.90	$0.60	**-$0.15**	-3%
20	typical (30%, Ask)	$0.27	$1.37	$0.60	**+$2.26**	50%
20	worst (100%, large builds)	$1.16	$5.80	$0.60	**-$3.05**	-68%

With typical use the plan stays healthy even at 20 free users per paying user. Only if every user — paying and free — spends their whole allowance on large builds does the plan lose money (at 10 or more free users per paying user). That is unlikely, but the levers below protect against it.

Levers that raise the margin without raising prices

- **Move Ask AI to a cheaper model.** It is the most-used action and runs on Claude Sonnet 4.6, the most expensive model we use. A cheaper model could cut its cost 3-5x; at 20 credits the margin on questions would grow accordingly.
- **Give Pro's promotional 1,000 credits for the first 3 months only.** Permanently, it doubles Pro's worst-case AI cost.
- **Free plan: 300 credits instead of 500**, or keep 500 but make Deep build Pro/Premium only. 300 credits cap a free user at $0.17 of AI a month.
- **Fill the server.** One VPS serves every user today; server cost per user falls with every new user until it needs upgrading.

9. App store fees

If subscriptions are sold inside the iOS or Android app, Apple and Google take 15% (small-business rate, under $1M a year; 30% above it) instead of Stripe's fee. Every plan stays profitable, with a higher AI share:

Plan (monthly, launch)	Through Stripe	Through app store (15%)	Through app store (30%)	AI worst case	AI share at 30% fee
Pro $4.99	$4.51	$4.24	$3.49	$1.16	33%
Premium $9.99	$9.33	$8.49	$6.99	$2.90	41%

Selling through the web (Stripe) keeps the most; if in-app purchase is required, the 15% rate is the one to plan for.

10. Features not priced yet

The plans list Study Studio features that are not built or not measured yet. Each needs a benchmark (run it at least 10 times on real material, as the draft rightly says) before it gets a price. The same 30% rule then applies.

Feature	Plan	How to charge
Notes Simplifier	Free	Per run, like a small build

Feature	Plan	How to charge
Teach Me / Adaptive Quiz Me	Free (limited) / Pro	Per AI call (each generated answer or question), like Ask AI
Standard mock exam	Pro	Per exam generated
Advanced mock exam	Premium	Per exam generated
Oral exam / Scenario modes	Premium	Per session with a cap on AI turns, or per AI turn — a long conversation under one flat price could exceed the 30% ceiling
Deeper weakness analysis	Premium	Per analysis
Heavier document processing	Premium	Upload limits, not credits — ingest has no per-call AI cost

Rule for session features: never sell an open-ended conversation for one flat price. Either charge per AI turn or cap the number of turns in a session so its largest possible cost is known.

11. Running it

- **Set the prices** in `system_settings` (keys in section 6). Changes apply within a minute.
- **Review monthly.** Every AI call is logged with its tokens and real cost (`llm_usage`). Once a month, compare the average real cost per action with its credit price; adjust if any action's AI share passes 30%.
- **Re-check when a model changes.** Model prices on OpenRouter change often, usually downwards; a model switch is the moment to re-run the benchmark.
- **Track free-user cost** as a separate line: AI used by free accounts per month, against the number of paying users.

12. Assumptions

- AI costs from one benchmark run per configuration on 8 October 2026 (AI Cost Sheet). Costs vary with the material; the "large" figures cover the biggest build possible today.
- Payment fees: Stripe 2.9% + $0.30 per charge, plus Stripe Billing 0.7% on subscriptions. Taxes and refunds not included.
- Plans, prices and allowances as in the Finance draft: Free 500 credits; Pro $4.99/month or $49.99/year, 1,000 + 1,000 promotional credits; Premium $9.99/month or $99.99/year, 5,000 credits; later prices $9.99 and $19.99 a month.
- Server cost ($60/month, 100 paying users) is a placeholder for illustration only.
- Study Studio only. OctoNotes, including transcription, is priced separately.